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Motor and pump system

Abstract

In a motor, a wire terminal, to which an end at an end of winding of a wire is connected, is held by an insulator that is held by a stator core. The wire terminal comprises: a substrate connection section; plural legs held by a split insulator; a plate section, a thickness direction of which faces a radial direction, between the legs and the substrate connection section; and a wire connection section, a portion of which projected from the plate section is bent to hold the wire on an inner side. An elastic section is a meandering section meandering to turn back in a circumferential direction between the plate section and the substrate connection section, and extends from an end on another side in the circumferential direction of the plate section toward the substrate connection section.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/561,630, filed on Dec. 23, 2021. The prior application Ser. No. 17/561,630, filed on Dec. 23, 2021 claims priority under 35 U.S.C. § 119 to Japanese Application No. 2020-216598 filed Dec. 25, 2020. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Field of the Invention

(1) At least an embodiment of the present invention relates to a motor in which an end of a wire extending from a coil is connected to a wire terminal, and to a pump system.

Description of the Related Documents

(2) For a motor having: a stator core in which plural salient poles are arranged in a circumferential direction with a motor axis being a center; an insulator that is held by the stator core; and a coil that is wound around the salient poles via the insulator, a structure of connecting an end of a wire that is drawn from the coil to a wire terminal is adopted (see Japanese Patent Application Publication No. 2017-103913). In the motor described in Japanese Patent Application Publication No. 2017-103913, the wire terminal has: a pin-shaped substrate connection section to which a substrate is connected; plural legs held by the insulator; a plate section that faces in a radial direction between the legs and the substrate connection section; and a wire connection section in which a portion projected from the plate section is bent to hold the wire on an inner side.

(3) In the technique described in Japanese Patent Application Publication No. 2017-103913, the substrate connection section is in the pin shape that extends linearly. Thus, when the substrate connection section is inserted in a hole of the substrate to connect the substrate connection section and a land of the substrate by soldering, stress may be applied to the substrate, which damages a circuit. In addition, in the case where heat generated by the coil is transferred to the wire terminal to cause thermal expansion of the wire terminal at the time of driving the motor, the substrate may be bent, which damages the circuit. Furthermore, when the stator is resin-sealed, a portion of the wire terminal is also resin-sealed. Thus, when the wire terminal expands thermally due to the heat generated during resin-sealing, the substrate may be bent, which damages the circuit.

(4) At least an embodiment of the present invention provides a motor and a pump system capable of suppressing transmission of stress from a wire terminal to a substrate.

SUMMARY

(5) A motor according to at least an embodiment of the present invention includes: a stator core in which plural salient poles are arranged in a circumferential direction with a motor axis being a center; an insulator that is held by the stator core; a coil that is wound around the salient pole via the insulator; a wire terminal that is held by the insulator and to which a wire extending from the coil is connected; and a substrate that is connected to the wire terminal. The wire terminal has: a substrate connection section to which the substrate is connected; plural legs that are held by the insulator; a plate section, a thickness direction of which faces a radial direction, between the leg and the substrate connection section; an elastic section that is elastically deformable between the plate section and the substrate connection section; and a wire connection section in which a portion projected from the plate section is bent to hold the wire on an inner side.

(6) In at least an embodiment of the present invention, since the elastic section is provided between the substrate connection section and the leg of the wire terminal, it is possible to absorb stress by the elastic section. Accordingly, it is possible to suppress transmission of the stress from the wire terminal to the substrate by the elastic section.

(7) At least an embodiment of the present invention can adopt such an aspect that the stator core, the insulator, the coil, and a portion of the wire terminal from the leg to the wire connection section are covered with a resin-sealed member. Even when such an aspect is adopted, stress generated by thermal expansion of the wire terminal is absorbed by the elastic section. Therefore, it is possible to suppress the transmission of the stress to the substrate.

(8) At least an embodiment of the present invention can adopt such an aspect that an end of the plate section on the side of the elastic section is exposed from the resin-sealed member. According to such an aspect, it is possible to suppress the elastic section from being covered with the resin-sealed member.

- (9) At least an embodiment of the present invention can adopt such an aspect that the plural legs include: a first leg that is projected, on one side in the circumferential direction of the wire connection section, from the plate section toward the insulator; and a second leg that is projected, on another side in the circumferential direction of the wire connection section, from the plate section toward the insulator, that the insulator is provided with a first hole into which the first leg is fitted and a second hole into which the second leg is fitted, that the first hole is a press-fitting hole into which the first leg is fitted, and that the second hole is a guide hole into which the second leg is fitted. According to such an aspect, since the second hole into which the other leg of the two legs is fitted is a guiding hole, it is easier to attach a common terminal to the insulator than a case where both of the first and second legs are press-fitted.
- (10) At least an embodiment of the present invention can adopt such an aspect that each of the first leg and the second leg is a square bar in a rectangular cross section, the first hole is a round hole in a circular cross section, and the second hole is a square hole in a rectangular cross section.
- (11) At least an embodiment of the present invention can adopt such an aspect that the second leg is longer than the first leg. According to such an aspect, since the second leg that is fitted into the guiding second hole is long, the common terminal can easily be guided. Therefore, the common terminal can easily be attached to the insulator.
- (12) At least an embodiment of the present invention can adopt such an aspect that the elastic section is a meandering section that meanders between the plate section and the substrate connection section, and, when seen in a direction of the motor axis, the meandering section does not overlap the first leg. According to such an aspect, when the end of the plate section on an opposite side of the insulator is pressed to press-fit the first leg into the first hole, the meandering section is unlikely to interfere with press-fitting.
- (13) At least an embodiment of the present invention can adopt such an aspect that, when seen in the direction of the motor axis, the elastic section meanders within a range that overlaps the plate section in the circumferential direction.
- (14) At least an embodiment of the present invention can adopt such an aspect that the elastic section has a pin shape that has a longer length dimension than the first leg and extends linearly toward the substrate connection section.
- (15) At least an embodiment of the present invention can adopt such an aspect that the elastic section extends from an end of the plate section on the other side in the circumferential direction toward the substrate connection section.
- (16) At least an embodiment of the present invention can adopt such an aspect that the wire connection section is electrically connected to the wire by fusing work.
- (17) The motor according to at least an embodiment of the present invention can be used in a pump system. In this case, the pump system is provided with an impeller that is rotationally driven by the motor.
- (18) In at least an embodiment of the present invention, since the elastic section is provided between the substrate connection section and the leg of the wire terminal, it is possible to absorb the stress by the elastic section. Therefore, it is possible to suppress the transmission of the stress from the wire terminal to the substrate by the elastic section.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will now be described, by way of example only, with reference to the accompanying drawings which are meant to be exemplary, not limiting, and wherein like elements are numbered alike in several figures, in which:
- (2) FIG. 1 is a perspective view illustrating an aspect of a pump system and a motor to which at

least an embodiment of the present invention is applied;

- (3) FIG. 2 is a vertical cross-sectional view of the pump system and the motor illustrated in FIG. 1;
- (4) FIG. 3 is an exploded perspective view illustrating a state where a cover is removed from the pump system illustrated in FIG. 1;
- (5) FIG. 4 is an exploded perspective view illustrating a state where a substrate is removed from the state illustrated in FIG. 3;
- (6) FIG. 5 is an exploded perspective view illustrating a state where a housing and a stator are separated from each other in the motor illustrated in FIG. 1;
- (7) FIG. 6 is a perspective view in which a portion around a wire terminal illustrated in FIG. 5 is seen from an inner side in a radial direction;
- (8) FIG. 7 is a perspective view in which the wire terminal illustrated in FIG. 5 is seen from the inner side in the radial direction;
- (9) FIG. 8 is a perspective view in which a portion around a common terminal illustrated in FIG. 5 is seen from the inner side in the radial direction;
- (10) FIG. 9 is a perspective view in which a first modified embodiment of the wire terminal illustrated in FIG. 7 is seen from the inner side in the radial direction; and
- (11) FIG. 10 is a perspective view in which a second modified embodiment of the wire terminal illustrated in FIG. 7 is seen from the inner side in the radial direction.

DETAILED DESCRIPTION

(12) A description will hereinafter be made on a motor and a pump system according to an embodiment of the present invention with reference to the drawings. In the following description, a motor axis L direction means a direction in which a motor axis L extends, a radial direction on an inner side in the radial direction and an outer side in the radial direction means a radial direction with the motor axis L being a center, and a circumferential direction means a rotational direction with the motor axis L being a center.

(13) Overall Configuration

(14) FIG. 1 is a perspective view illustrating an aspect of a pump system **1** and a motor **10** to which at least an embodiment of the present invention is applied. FIG. 2 is a vertical cross-sectional view of the pump system **1** and the motor **10** illustrated in FIG. 1. In FIG. 1 and FIG. 2, the pump system **1** has: a case **2** that includes a suction port **21a** and a discharge port **22a**; the motor **10** that is arranged on one side **L1** in the motor axis L direction of the case **2**; and an impeller **25** that is arranged in a pump chamber **20** inside the case **2**. The impeller **25** is rotationally driven around the motor axis L by the motor **10**. The motor **10** includes: a cylindrical stator **3**; a rotor **4** that is arranged in the stator **3**; a housing **6** that is made of resin and covers the stator **3**; and a support shaft **5** that has a round rod shape and supports the rotor **4** in a manner to allow rotation thereof. The support shaft **5** is made of metal or ceramic. In the pump system **1** of this form, a fluid is a liquid, and the pump system **1** is used under such a condition that an environmental temperature and a fluid temperature are likely to change.

(15) The case **2** constitutes a wall surface **23** on another side **L2** in the motor axis L direction of the pump chamber **20** and a side wall **29** that extends in the circumferential direction. The case **2** includes: a suction pipe **21** that extends along the motor axis L; and a discharge pipe **22** that extends in a direction orthogonal to the motor axis L. Each of the suction pipe **21** and the discharge pipe **22** has the suction port **21a** and the discharge port **22a** at an end. The suction pipe **21** is concentrically provided to the motor axis L.

(16) In the motor **10**, the stator **3** has: a stator core **31**; insulators **32**, **33** that are held by the stator core **31**; and a coil **35** that is wound on the stator core **31** via the insulators **32**, **33**.

(17) The rotor **4** includes a cylindrical section **40** that extends from a position on an inner side in the radial direction opposing the stator **3** toward the pump chamber **20** along the motor axis L, and the cylindrical section **40** is opened in the pump chamber **20**. A cylindrical magnet **47** is held on an outer circumferential surface of the cylindrical section **40** in a manner to oppose the stator **3** at the

position on the inner side in the inner radial direction thereof. The magnet **47** is a neodymium bonded magnet, for example.

(18) In the rotor **4**, a disc-shaped flange section **45** is formed at an end of the other side **L2** in the motor axis **L** direction of the cylindrical section **40**, and a disc **26** is coupled to the flange section **45** from the other side **L2** in the motor axis **L** direction. A central hole **260** is formed at a center of the disc **26**. Plural vane sections **261**, each of which extends outward in the radial direction while being curved arcuately around the central hole **260** are formed at equal angular intervals on a surface of the disc **26** opposing the flange section **45**, and the disc **26** is fixed to the flange section **45** via the vane sections **261**. Accordingly, the flange section **45** and the disc **26** constitute the impeller **25** that is connected to the cylindrical section **40** of the rotor **4**. In this embodiment, the disc **26** is inclined such that a portion thereof on the outer side in the radial direction is located on the flange section **45** side from the inner side in the radial direction.

(19) In the rotor **4**, a cylindrical radial bearing **11** is held on the inner side in the radial direction of the cylindrical section **40** by a method such as caulking, and the rotor **4** is rotatably supported on the support shaft **5** via the radial bearing **11**. A first end **51** on the one side **L1** in the motor axis **L** direction of the support shaft **5** is held in a shaft hole **65** that is formed in a bottom wall **63** of the housing **6**. The case **2** is formed with a receiving section **280** that opposes, on the pump chamber **20** side, a second end **52** on the pump chamber **20** side of the support shaft **5** and limits a movable range of the support shaft **5** to the pump chamber **20** side. The case **2** includes three support sections **27**, each of which extends from an inner circumferential surface of the suction pipe **21** to the motor **10** side. A cylindrical section **28**, in which the support shaft **5** is located, is formed at an end of the support section **27**, and the receiving section **280** is constructed of a bottom section on the other side **L2** in the motor axis **L** direction of the cylindrical section **28**. A circular thrust bearing **12** is mounted on the second end **52** of the support shaft **5**. The thrust bearing **12** is arranged between the radial bearing **11** and the cylindrical section **28**. Here, the first end **51** and the shaft hole **65** are at least partially formed to have a D-shaped cross section, and the second end **52** of the support shaft **5** and a hole of the thrust bearing **12** are formed to have a D-shaped cross section. In this way, rotation of the support shaft **5** and the thrust bearing **12** is inhibited.

(20) The housing **6** is a partition wall member having: a first partition wall section **61** that opposes the wall surface **23** of the pump chamber **20**; and a second partition wall section **62** that is interposed between the stator **3** and the magnet **47**. The housing **6** also has a cylindrical trunk section **66** that covers the stator **3** from the outer side in the radial direction. Accordingly, the housing **6** is a resin-sealed member **60** that covers the stator **3** from both sides in the radial direction and from both sides in the motor axis **L** direction, and is a resin portion at the time of insertion molding of the stator **3** by polyphenylene sulfide (PPS) or the like.

(21) Detailed Configuration of Motor **10**

(22) FIG. **3** is an exploded perspective view illustrating a state where the cover **18** is removed from the pump system **1** illustrated in FIG. **1**. FIG. **4** is an exploded perspective view illustrating a state where a substrate **19** is removed from the state illustrated in FIG. **3**. FIG. **5** is an exploded perspective view illustrating a state where the housing **6** and the stator **3** are separated from each other in the motor **10** illustrated in FIG. **1**. In FIG. **3**, FIG. **4**, and FIG. **5**, the motor axis **L** direction is reversed vertically from that in FIG. **1** and FIG. **2**, and the one side **L1** in the motor axis **L** direction is set on an upper side in the drawings.

(23) As illustrated in FIG. **2**, FIG. **3**, and FIG. **4**, the cover **18** is fixed to an end **64** on the one side **L1** in the motor axis **L** direction of the housing **6** from the one side **L1** in the motor axis **L** direction, and the substrate **19** that is provided with a circuit controlling power supply to the coil **35**, and the like is arranged between the cover **18** and the bottom wall **63** of the housing **6**.

(24) The substrate **19** is fixed to the housing **6** by a first fixing section **97** using a first screw **91** and a second fixing section **98** using a second screw **92**. The housing **6** is formed with a first cylindrical section **67** that is projected from the bottom wall **63** toward the one side **L1** in the motor axis **L**

direction. The first fixing section **97** is constituted by fastening the first screw **91**, which is constructed of a tapping screw, to the first cylindrical section **67** via a notch **197** that is formed at an edge of the substrate **19**. The housing **6** is formed with a second cylindrical section **68** that is projected from the bottom wall **63** on an opposite side with the motor axis L being the center toward the one side L1 in the motor axis L direction. The second fixing section **98** is constituted by fastening the second screw **92**, which is constructed of a tapping screw, to the second cylindrical section **68** via a notch **198** that is formed at the edge of the substrate **19**.

(25) The substrate **19** is provided with: plural first connection sections **191**, to each of which a metal wire terminal **71** that passes through the bottom wall **63** of the housing **6** from the stator **3** and is projected to the one side L1 in the motor axis L direction is connected by soldering; and a second connection section **192**, to which a metal connector terminal **75** that is held by the housing **6** is connected by soldering. The substrate **19** is formed with a wire that electrically connects the second connection section **192** and each of the first connection sections **191** via a drive circuit and the like mounted on the substrate **19**, and the like.

(26) A cylindrical connector housing **69** is formed in the housing **6**, and an end of the connector terminal **75** is located in the connector housing **69**. Accordingly, when a connector is connected to the connector housing **69** and a signal or the like is supplied thereto, such a signal is input to the drive circuit via the connector terminal **75** and the second connection section **192**. As a result, a drive current that is generated by the drive circuit is supplied to each of the coils **35** via the first connection section **191** and the wire terminal **71**. Consequently, the rotor **4** rotates about the motor axis L. In this way, the impeller **25** rotates in the pump chamber **20** and a negative pressure is generated in the pump chamber **20**. Thus, the fluid is suctioned into the pump chamber **20** from the suction pipe **21** and is discharged from the discharge pipe **22**.

(27) As illustrated in FIG. 2 and FIG. 5, in the stator **3**, the stator core **31** includes: a circular section **311** that extends in a circular shape; and plural salient poles **312**, each of which is projected inward in the radial direction from the circular section **311**. The salient poles **312** are arranged at a constant pitch in the circumferential direction. The stator core **31** is a laminated core that is formed by laminating thin magnetic plates, each of which is made of a magnetic material. A recessed section **315** that extends in the motor axis L direction is formed on an outer circumferential surface of the circular section **311** in a manner to correspond to the plural salient poles **312**. In this embodiment, the stator core **31** is formed by bending a linearly extending member into a circular shape and then welding ends of the circular section **311** together. Accordingly, the stator core **31** includes a weld section **310** that connects circumferentially extending portions of the circular section **311** at one position in the circumferential direction.

(28) The insulators **32**, **33** overlap the stator core **31** from both sides in the motor axis L direction and cover each of the plural salient poles **312**. In this embodiment, the insulators **32**, **33** respectively include plural split insulators **320**, **330**, each of which is split in a manner to correspond to respective one of the plural salient poles **312**. The plural split insulators **320**, **330** respectively include: outer circumferential portions **321**, **331**, each of which overlaps the circular section **311** of the stator core **31** in the motor axis L direction; inner circumferential portions **322**, **332**, each of which is projected in the motor axis L direction at a radially inner end of the salient pole **312**; and a cylinder forming portion (not illustrated) that respectively connects the outer circumferential portions **321**, **331** and the inner circumferential portions **322**, **332**. The coil **35** is wound around the salient poles **312** via the cylinder forming portion.

(29) The motor **10** is a three-phase motor. Accordingly, in the plural coils **35**, a first phase coil **35(U)** including a U-phase coil, a second phase coil **35(V)** including a V-phase coil, and a third phase coil **35(W)** including a W-phase coil are arranged in this order. In this embodiment, three each of the first phase coils **35(U)**, the second phase coils **35(V)**, and the third phase coils **35(W)** are arranged, and the total number of the coils **35** is nine. Accordingly, a total of the nine split insulators **320** is arranged, and the nine split insulators **320** have the same configuration.

(30) In this embodiment, a guide groove 335 for the wire 350 at the time of winding the coil 35 is formed on an outer surface of the outer circumferential portion 321 in each of the nine split insulators 330, and the first phase coil 35(U) is formed of a single wire. Accordingly, the three first phase coils 35(U) are electrically connected in series. The same applies to the second phase coil 35(V) and the third phase coil 35(W).

(31) In addition, each of the wire terminals 71 is held by the outer circumferential portion 321 of respective one of the split insulator 320 that corresponds to one of the first phase coils 35(U), the split insulator 320 that corresponds to one of the second phase coils 35(V), and the split insulator 320 that corresponds to one of the third phase coils 35(W) of the nine split insulators 320. One end 351 of the wire 350 that constitutes each of the three coils 35 connected in series is connected to each of the three wire terminals 71, and another end 352 is electrically connected to a metal common terminal 72 that is held by the outer circumferential portion 321 of the other split insulator 320. In this embodiment, the other end 352 is an end at the beginning of winding, and the one end 351 is an end at the end of winding.

(32) The wire terminal 71 is projected from the split insulator 320 toward the one side L1 in the motor axis L direction, and a substrate connection section 711 as a tip of the wire terminal 71 is connected to the substrate 19 illustrated in FIG. 2, FIG. 3, and FIG. 4.

(33) Configuration of Wire Terminal 71

(34) FIG. 6 is a perspective view in which a portion around the wire terminal 71 illustrated in FIG. 5 is seen from the inner side in the radial direction. FIG. 7 is a perspective view in which the wire terminal 71 illustrated in FIG. 5 is seen from the inner side in the radial direction. Similar to FIG. 3 and the like, also in FIG. 6 and FIG. 7, the one side L1 in the motor axis L direction is set as an upper side.

(35) As illustrated in FIG. 5, the wire terminal 71, to which the winding end 351 of the wire 350 extending from the coil 35, is held by each of the three split insulators 320 with different corresponding phases among the split insulators 320 not provided with the common terminal 72.

(36) As illustrated in FIG. 6 and FIG. 7, the wire terminal 71 has: the substrate connection section 711; plural legs 715 held by the split insulator 320; a plate section 710, a thickness direction of which faces the radial direction, between the legs 715 and the substrate connection section 711; and a wire connection section 718, a portion of which projected from the plate section 710 is bent to hold the wire 350 on an inner side. The wire connection section 718 has a portion that is projected from an end 710a on the leg 715 side of the plate section 710 and is bent upward so as to hold the wire 350 on the inner side. Accordingly, the wire connection section 718 holds the end 351 of the wire 350 by fusing work and is electrically connected to the end 351 of the wire 350. The fusing work is thermal caulking processing that connects the wire 350 and the wire connection section 718 by using electrical resistance.

(37) In this embodiment, the wire terminal 71 has an elastic section 714 that is elastically deformable between the plate section 710 and the substrate connection section 711. Here, as illustrated in FIG. 4, the stator core 31, the insulator 32, the coil 35, and a portion of the wire terminal 71 from the leg 715 to the wire connection section 718 are covered with a resin-sealed member 60. However, the substrate connection section 711 and the elastic section 714 are projected from the bottom wall 63 of the resin-sealed member 60 to the one side L1 in the motor axis L direction and are exposed. More specifically, a large portion of the plate section 710 is covered with the resin-sealed member 60. However, an end 710b on the elastic section 714 side of the plate section 710 is exposed from the bottom wall 63 of the resin-sealed member 60. Accordingly, the substrate connection section 711 and the elastic section 714 are projected from the bottom wall 63 of the resin-sealed member 60 to the one side L1 in the motor axis L direction and are exposed.

(38) In this embodiment, of the two legs 715, one of the legs 715 is a first leg 716, and the other leg 715 is a second leg 717 that is provided on another side CCW in the circumferential direction of the first leg 716. Corresponding to such a configuration, the split insulator 320 is provided with: a first

hole **326** into which the first leg **716** is fitted; and a second hole **327** into which the second leg **717** is fitted on the other side CCW in the circumferential direction of the first leg **716**.

(39) In this embodiment, the first hole **326** is a press-fitting hole into which the first leg **716** is fitted, and the second hole **327** is a guide hole into which the second leg **717** is fitted. Here, each of the two legs **715** is a square bar in a rectangular cross section and has an equal thickness. However, the first hole **326** is a round hole in a circular cross section, and the second hole **327** is a square hole in a rectangular cross section. Accordingly, the second hole **327** can constitute the guide hole, and the first hole **326** can constitute the press-fitting hole. Here, the first hole **326** is opened at a bottom of a recessed section **329** that is formed by cutting out a surface on the one side **L1** in the motor axis **L** direction of the split insulator **320** and a radially inner surface.

(40) In this embodiment, the elastic section **714** is a meandering section **712** that meanders to turn back in the circumferential direction at a position between the plate section **710** and the substrate connection section **711**. Here, the elastic section **714** extends from an end **710c** on the other side CCW in the circumferential direction of the plate section **710** toward the substrate connection section **711**. When seen in the motor axis **L** direction, the meandering section **712** does not overlap the first leg **716** that is used to press-fit. Accordingly, when the end **710a** of the plate section **710** on the opposite side of the split insulator **320** is pressed to press-fit the first leg **716** into the first hole **326**, the meandering section **712** is unlikely to interfere with press-fitting. In addition, when seen in the motor axis **L** direction, the elastic section **714** meanders within a range that overlaps the plate section **710** in the circumferential direction and does not bulge from the plate section **710** in the circumferential direction. Thus, it is possible to reduce a circumferential dimension of the wire terminal **71**.

(41) As described above, in this embodiment, since the elastic section **714** is provided between the substrate connection section **711** and the leg **715** of the wire terminal **71**, it is possible to absorb stress by the elastic section **714**. Accordingly, it is possible to suppress transmission of the stress from the wire terminal **71** to the substrate **19** by the elastic section **714**. For example, it is possible to suppress the stress from being applied to the substrate **19** and damaging the circuit at the time when the substrate connection section **711** is inserted in the hole of the substrate **19** and the substrate connection section **711** is connected to the land of the substrate **19** by soldering. In addition, also in the case where heat generated by the coil **35** is transferred to the wire terminal **71** to cause thermal expansion of the wire terminal **71** at the time of driving the motor **10**, it is possible to suppress the substrate **19** from being bent, which damages the circuit. Furthermore, when the stator **3** is resin-sealed, a portion of the wire terminal **71** is also resin-sealed. Thus, even when the wire terminal **71** expands thermally due to the heat generated during resin-sealing, it is possible to suppress the substrate **19** from being bent, which damages the circuit.

(42) Configuration of Common Terminal **72**

(43) FIG. **8** is a perspective view in which a portion around the wire common terminal **72** illustrated in FIG. **5** is seen from the inner side in the radial direction. Similar to FIG. **3** and the like, also in FIG. **8**, the one side **L1** in the motor axis **L** direction is set as the upper side.

(44) As illustrated in FIG. **5**, a common terminal **72**, to which the ends **352** at the beginning of winding of the wires **350** in the number corresponding to the number of the phases of the coils **35** are connected, is held by the split insulator **320(W)** of the split insulators **320** not provided with the wire terminal **71**.

(45) In this embodiment, the common terminal **72** includes: a plate section **720** that extends in the circumferential direction; plural wire connection sections **721** in which portions projected from plural positions in the circumferential direction of the plate section **720** are each bent to hold the end **352** of the wire **350** on the inner side; and a leg **725** that is projected from the plate section **720** toward the split insulator **320** at a position between one adjacent pair of the wire connection sections **721** among the plural wire connection sections **721**. The leg **725** is held by the split insulator **320**. The leg **725** is provided only between the adjacent wire connection sections **721** of

the plural wire connection sections **721**. Accordingly, since the number of the legs **725** is small, the plate section **720** has a short circumferential dimension. Thus, the plate section **720** is formed in a flat plate shape that extends linearly along the circumferential direction.

(46) In addition, the plural wire connection sections **721** are each bent to hold the end **352** of the wire **350** on the inner side, and each of the wire connection sections **721** holds the end **352** of the wire **350** by the fusing work and is electrically connected to the end **352** of the wire **350**.

(47) In this embodiment, since the number of the phases is three, the three wire connection sections **721** are provided, and the two legs **725** are provided. Of the two legs **725**, one of the legs **725** is a first leg **726**, and the other leg **725** is a second leg **727** that is provided on the other side CCW in the circumferential direction of the first leg **726**. Corresponding to such a configuration, similar to the split insulator **320** that holds the wire terminal **71**, the split insulator **320** is provided with: the first hole **326** into which the first leg **726** is fitted; and the second hole **327** into which the second leg **727** is fitted on the other side CCW in the circumferential direction of the first leg **726**.

(48) In this embodiment, the first hole **326** is the press-fitting hole into which the first leg **726** is fitted, and the second hole **327** is the guide hole into which the second leg **727** is fitted. Here, each of the two legs **725** is the square bar in the rectangular cross section and has the equal thickness. However, the first hole **326** is the round hole in the circular cross section, and the second hole **327** is the square hole in the rectangular cross section. Accordingly, the second hole **327** can constitute the guide hole, and the first hole **326** can constitute the press-fitting hole.

(49) The three wire connection sections **721** include: a first wire connection section **721(U)**; a second wire connection section **721(V)** that is provided on the other side CCW in the circumferential direction of the first wire connection section **721(U)**; and a third wire connection section **721(W)** that is provided between the first wire connection section **721(U)** and the second wire connection section **721(V)**. The end **352** of the wire **350** that extends from the first phase coil **35(U)** reaches the first wire connection section **721(U)** from one side CW in the circumferential direction. The end **352** of the wire **350** that extends from the second phase coil **35(V)** reaches the second wire connection section **721(V)** from the other side CCW in the circumferential direction. The end **352** of the wire **350** that extends from the third phase coil **35(W)** reaches the third wire connection section **721(W)** from the one side CW in the circumferential direction. Here, the split insulator **320** is provided with a guiding recessed section **329** between the first wire connection section **721(U)** and the third wire connection section **721(W)**, and the guiding recessed section **329** guides the end **352** of the wire **350**, which extends from the third phase coil **35(W)**, to the third wire connection section **721(W)** to the inner side in the radial direction. Accordingly, even in the case where each of the end **352** of the wire **350** extending from the first phase coil **35(U)** and the end **352** of the wire **350** extending from the third phase coil **35(W)** reaches from the one side CW in the circumferential direction, the end **352** of the wire **350** extending from the third phase coil **35(W)** can be guided to the third wire connection section **721(W)** through the guiding recessed section **329**, and thus can easily be connected to the common terminal **72**.

(50) As described above, in the motor **10** and the pump system **1** of this embodiment, the common terminal **72** is provided with the single leg **725** between one pair of the adjacent wire connection sections **721**. The leg **725** is provided only between the adjacent wire connection sections **721** of the plural wire connection sections **721**, and the leg **725** is not provided between both ends of the plate section **720**. Accordingly, since the common terminal **72** has a short circumferential dimension, it is possible to reduce cost of the common terminal **72**. In addition, since the number of the legs **725** is small, the common terminal **72** can easily be attached to the insulator **32**.

(51) Since the circumferential dimension of the common terminal **72** is short, it is possible to realize a structure of holding the common terminal **72** by the single split insulator **320**. Thus, relative positional accuracy between the first hole **326** and the second hole **327**, into which the leg **725** is fitted, is high. Therefore, the common terminal **72** can easily be attached. In addition, since the circumferential dimension of the common terminal **72** is short, the plate section **720** of the

common terminal 72 can have the flat plate shape, and a process of curving the plate section 720 is unnecessary. Therefore, the cost of the common terminal 72 can be reduced.

(52) In the insulator 32, the first hole 326 is the press-fitting hole into which the first leg 726 is fitted, and the second hole 327 is the guide hole into which the second leg 727 is fitted.

Accordingly, the first leg 726 can be press-fitted into the first hole 326 while the second leg 727 and the second hole 327 guide the common terminal 72. Thus, compared to a case where both of the two legs are press-fitted, the common terminal 72 can easily be attached to the insulator 32.

First Modified Embodiment of Wire Terminal 71

(53) FIG. 9 is a perspective view in which a first modified embodiment of the wire terminal 71 illustrated in FIG. 7 is seen from the inner side in the radial direction. Since a basic configuration in this modified embodiment is similar to that in the above embodiment, common portions will be denoted by the same reference numerals, and a description thereon will not be made.

(54) In the wire terminal 71 that has been described with reference to FIG. 7 and the like, the first leg 716 and the second leg 717 have the equal length. However, in this modified embodiment, as illustrated in FIG. 9, the guiding second leg 717 is longer than the press-fitting first leg 716.

Accordingly, the common terminal 72 can easily be guided when the first leg 716 is press-fitted into the first hole 326 while the second leg 717 and the second hole 327 guide the wire terminal 71.

Second Modified Embodiment of Wire Terminal 71

(55) FIG. 10 is a perspective view in which a second modified embodiment of the wire terminal 71 illustrated in FIG. 7 is seen from the inner side in the radial direction. Since a basic configuration in this modified embodiment is similar to that in the above embodiment, common portions will be denoted by the same reference numerals, and a description thereon will not be made.

(56) In the wire terminal 71 that has been described with reference to FIG. 7 and the like, the elastic section 714 is the meandering section 712. However, in this modified embodiment, as illustrated in FIG. 10, the elastic section 714 is a pin-shaped shaft section 713 that has a longer dimension than the first leg 716 and extends linearly toward the substrate connection section 711. For example, the shaft section 713 that constitutes the elastic section 714 is 1.2 to 1.5 times longer than the first leg 716. Accordingly, the elastic section 714 has appropriate elasticity.

(57) Here, the elastic section 714 extends from the end 710c on the other side CCW in the circumferential direction of the plate section 710 toward the substrate connection section 711. When seen in the motor axis L direction, the elastic section 714 does not overlap the first leg 716 that is used to press-fit. Accordingly, when the end 710a of the plate section 710 on the opposite side of the split insulator 320 is pressed to press-fit the first leg 716 into the first hole 326, the elastic section 714 is unlikely to interfere with press-fitting.

OTHER EMBODIMENTS

(58) The motor 10 that is used in the pump system 1 has been exemplified in the above embodiment. However, at least an embodiment of the present invention may be applied to a motor that is mounted on another equipment.

Claims

1. A motor comprising: a stator core in which plural salient poles are arranged in a circumferential direction with a motor axis being a center; an insulator that is held by the stator core; a coil that is wound around the salient pole via the insulator; a wire terminal that is held by the insulator and to which a wire extending from the coil is connected; and a substrate that is connected to the wire terminal, wherein the wire terminal comprises: a substrate connection section to which the substrate is connected; at least one leg that is held by the insulator; a plate section, a thickness direction of which faces a radial direction, between the leg and the substrate connection section; a section for suppressing stress transmission that has a first terminal and a second terminal, wherein the first terminal is connected to the plate section, and the second terminal is connected to the substrate

connection section; and a wire connection section in which a portion projected from the plate section is bent to hold the wire on an inner side.

2. The motor according to claim 1, wherein the stator core, the insulator, the coil, and a portion of the wire terminal from the leg to the wire connection section are covered with a resin-sealed member.

3. The motor according to claim 2, wherein an end of the plate section on the side of the section for suppressing stress transmission is exposed from the resin-sealed member.

4. The motor according to claim 1, wherein the at least one leg comprises: a first leg that is projected, on one side in the circumferential direction of the wire connection section, from the plate section toward the insulator; and a second leg that is projected, on another side in the circumferential direction of the wire connection section, from the plate section toward the insulator, and the insulator is provided with a first hole into which the first leg is fitted and a second hole into which the second leg is fitted.

5. The motor according to claim 4, wherein the first hole is a press-fitting hole into which the first leg is fitted, and the second hole is a guide hole into which the second leg is fitted.

6. The motor according to claim 5, wherein each of the first leg and the second leg is a square bar in a rectangular cross section, the first hole is a round hole in a circular cross section, and the second hole is a square hole in a rectangular cross section.

7. The motor according to claim 5, wherein the second leg is longer than the first leg.

8. The motor according to claim 5, wherein the section for suppressing stress transmission is a meandering section that meanders between the plate section and the substrate connection section, and when seen in a direction of the motor axis, the meandering section does not overlap the first leg.

9. The motor according to claim 8, wherein when seen in the direction of the motor axis, the section for suppressing stress transmission meanders within a range that overlaps the plate section in the circumferential direction.

10. The motor according to claim 5, wherein the section for suppressing stress transmission comprises a pin shape that comprises a longer length dimension than the first leg and extends linearly toward the substrate connection section.

11. The motor according to claim 1, wherein the section for suppressing stress transmission extends from an end of the plate section on the other side in the circumferential direction toward the substrate connection section.

12. The motor according to claim 1, wherein the wire connection section is electrically connected to the wire by fusing work.

13. The motor according to claim 1, wherein at least a portion of the leg extends on a same axis as the substrate connection section.

14. The motor according to claim 1, wherein a length of a portion of the insulator holding the leg is narrower than a width of the coil being wound.

15. The motor according to claim 1, wherein the section for suppressing stress transmission is an elastic portion that is elastically deformable between the plate section and the substrate connection section.

16. A pump system comprising: the motor according to claim 1, and further comprising: an impeller that is rotationally driven by the motor.

17. A motor comprising: a stator core in which plural salient poles are arranged in a circumferential direction with a motor axis being a center; an insulator that is held by the stator core; a coil that is wound around the salient pole via the insulator; a wire terminal that is held by the insulator and to which a wire extending from the coil is connected; and a substrate that is connected to the wire terminal, wherein the wire terminal comprises: a substrate connection section to which the substrate is connected; a leg that is held by the insulator; a plate section, a thickness direction of which faces a radial direction, between the leg and the substrate connection section; an elastic portion that is

elastically deformable and has a first terminal and a second terminal, wherein the first terminal is connected to the plate section, and the second terminal is connected to the substrate connection section; and a wire connection section in which a portion projected from the plate section is bent to hold the wire on an inner side.
